SG No. 218 T_{d}^4 $P\bar{4}3n$ [cubic]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `23.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `6b`, site symmetry: `222..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[13,14,15,16]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[17,18,19,20]`
6	$\left[0, 0, \frac{1}{2}\right]$	`[21,22,23,24]`

Table 3: Wyckoff site: `6c`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,4,18,19]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,3,17,20]`
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,8,14,15]`
4	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,13,16]`
5	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,12,22,23]`
6	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,21,24]`

Table 4: Wyckoff site: `6d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,4,18,19]`
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,3,17,20]`
3	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,7,13,16]`
5	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[9,12,22,23]`
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,11,21,24]`

Table 5: Wyckoff site: `8e`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,9]`
2	$[-x, -x, x]$	`[2,7,12]`
3	$[-x, x, -x]$	`[3,8,10]`

continued ...

Table 5

No.	position	mapping
4	$[x, -x, -x]$	`[4,6,11]`
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[13,17,21]`
6	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	`[14,19,24]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[15,20,22]`
8	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	`[16,18,23]`

Table 6: Wyckoff site: `12f`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4]`
2	$[-x, 0, 0]$	`[2,3]`
3	$[0, x, 0]$	`[5,8]`
4	$[0, -x, 0]$	`[6,7]`
5	$[0, 0, x]$	`[9,12]`
6	$[0, 0, -x]$	`[10,11]`
7	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[13,16]`
8	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[14,15]`
9	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,20]`
10	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	`[18,19]`
11	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	`[21,24]`
12	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	`[22,23]`

Table 7: Wyckoff site: `12g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	`[1,4]`
2	$\left[-x, \frac{1}{2}, 0\right]$	`[2,3]`
3	$\left[0, x, \frac{1}{2}\right]$	`[5,8]`
4	$\left[0, -x, \frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{2}, 0, x\right]$	`[9,12]`
6	$\left[\frac{1}{2}, 0, -x\right]$	`[10,11]`
7	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	`[13,16]`
8	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	`[14,15]`
9	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	`[17,20]`
10	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	`[18,19]`
11	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	`[21,24]`
12	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	`[22,23]`

Table 8: Wyckoff site: `12h`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,4]`
2	$\left[-x, 0, \frac{1}{2}\right]$	`[2,3]`

continued ...

Table 8

No.	position	mapping
3	$\left[\frac{1}{2}, x, 0\right]$	[5,8]
4	$\left[\frac{1}{2}, -x, 0\right]$	[6,7]
5	$\left[0, \frac{1}{2}, x\right]$	[9,12]
6	$\left[0, \frac{1}{2}, -x\right]$	[10,11]
7	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[13,16]
8	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[14,15]
9	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[17,20]
10	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[18,19]
11	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[21,24]
12	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[22,23]

Table 9: Wyckoff site: 24i, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-x, -y, z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[x, -y, -z]$	[4]
5	$[z, x, y]$	[5]
6	$[z, -x, -y]$	[6]
7	$[-z, -x, y]$	[7]
8	$[-z, x, -y]$	[8]
9	$[y, z, x]$	[9]
10	$[-y, z, -x]$	[10]
11	$[y, -z, -x]$	[11]
12	$[-y, -z, x]$	[12]
13	$\left[y+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[13]
14	$\left[\frac{1}{2}-y, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[14]
15	$\left[y+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-y, x+\frac{1}{2}, \frac{1}{2}-z\right]$	[16]
17	$\left[x+\frac{1}{2}, z+\frac{1}{2}, y+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-x, z+\frac{1}{2}, \frac{1}{2}-y\right]$	[18]
19	$\left[\frac{1}{2}-x, \frac{1}{2}-z, y+\frac{1}{2}\right]$	[19]
20	$\left[x+\frac{1}{2}, \frac{1}{2}-z, \frac{1}{2}-y\right]$	[20]
21	$\left[z+\frac{1}{2}, y+\frac{1}{2}, x+\frac{1}{2}\right]$	[21]
22	$\left[z+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-x\right]$	[22]
23	$\left[\frac{1}{2}-z, y+\frac{1}{2}, \frac{1}{2}-x\right]$	[23]
24	$\left[\frac{1}{2}-z, \frac{1}{2}-y, x+\frac{1}{2}\right]$	[24]